# Model

# 1 Model

## 1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \ldots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network. The generalized commuting cost from residence $i$ to workplace $j$ is

$$\tau_{ij} = \tau_{ij}(I_{ij},\, n_{ij}),$$

where $I_{ij}$ is transportation investment on link $(i, j)$ and $n_{ij}$ is the equilibrium commuter flow on that link. Investment reduces commuting costs and congestion raises them:

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0, \qquad \frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the commuting cost as

$$\tau_{ij} = \delta_{ij}^{\tau} \frac{n_{ij}^{\rho}}{I_{ij}^{\gamma}},$$

where $\delta_{ij}^{\tau} > 0$ captures geographic features of link $(i, j)$ and $\rho, \gamma > 0$.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

## 1.2 Households

A household chooses a residence location $i$ and a workplace $j$ to maximize utility. Preferences over the numaire good $Q_{ij}$ and residential floor space $h_{ij}^R$ are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha}\right)^{\alpha} \left(\frac{h_{ij}^R}{1-\alpha}\right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location $i$, $\alpha \in (0,1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each $(i,j)$ pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \le w_j - \phi(N_i, L_i),$$

where $q_i^R$ is the residential rent at location $i$, $w_j$ is the wage at workplace $j$. Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1-\alpha)Y_{ij}/q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i\,(w_j - \phi(N_i, L_i))}{\tau_{ij}\,(q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \le z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence $i$ and workplace $j$ is

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^{\epsilon}}{\Phi}, \qquad \Phi \equiv \sum_{i'} \Phi_{i'}, \qquad \Phi_i \equiv \sum_j T_i E_j \tilde{V}_{ij}^{\epsilon}, \tag{1}$$

and expected utility is

$$U = \Gamma\big(1 - \tfrac{1}{\epsilon}\big)\,\Phi^{1/\epsilon}. \tag{2}$$

Let $N$ denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location $i$ pins down the equilibrium rent:

$$q_i^R = \frac{(1-\alpha)\sum_j N\pi_{ij} Y_{ij}}{\bar{H}_i^R}. \tag{3}$$

## 1.3 Firms

At each location $j$, competitive firms produce a freely tradable good using labor $L_j$ and commercial floor space $H_j^F$:

$$Y_j = A_j L_j^{\gamma} (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0,1)$. Taking wages $w_j$ and commercial rents $q_j^F$ as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j}\right)^{1-\gamma}, \qquad q_j^F = (1-\gamma) A_j \left(\frac{L_j}{H_j^F}\right)^{\gamma}.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at $j$ gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses $L_i$ to denote employment at location $i$ acting as a workplace, which is this same quantity evaluated at $j = i$.

## 1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality $g \in G$ chooses a fee $\{t_i^F\}_{i \in S^g}$ levied on commercial land use at each location — an additional cost incurred whenever land is used for commercial rather than residential purposes, in the spirit of real-world commercial linkage fees that charge new commercial floor space to help offset the housing demand it generates. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher after-fee return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$q_i^R = q_i^F - t_i^F. \tag{4}$$

Together with the land identity $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$, the residential rent $q_i^R(\bar{H}_i^R)$ from (3) and the commercial rent $q_i^F(\bar{H}_i^F)$ from the firm's first-order condition — each strictly decreasing in its own floor-space argument — condition (4) pins down the equilibrium land split given the fee. Define

$$g(\bar{H}_i^R; t_i^F) \equiv q_i^R(\bar{H}_i^R) - q_i^F(\bar{H}_i - \bar{H}_i^R) + t_i^F.$$

Since $q_i^R$ is strictly decreasing in $\bar{H}_i^R$ and $q_i^F$ is strictly decreasing in $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ (so $-q_i^F(\bar{H}_i - \bar{H}_i^R)$ is itself strictly decreasing in $\bar{H}_i^R$), $g$ is strictly decreasing in $\bar{H}_i^R$, so $g(\bar{H}_i^R; t_i^F) = 0$ has a unique solution $\bar{H}_i^R(t_i^F)$ for every fee. By the implicit function theorem,

$$\frac{d\bar{H}_i^R}{dt_i^F} = -\frac{1}{g'(\bar{H}_i^R)} > 0, \qquad g'(\bar{H}_i^R) \equiv \frac{\partial q_i^R}{\partial \bar{H}_i^R} + \frac{\partial q_i^F}{\partial \bar{H}_i^F} < 0, \tag{5}$$

so a higher commercial-use fee raises equilibrium residential floor space: taxing commercial use pushes land toward residential. No supplier or developer is introduced — (4) is a no-arbitrage condition on a competitive land market, using only the rent functions the model already has, and total land $\bar{H}_i$ remains fixed.

The fee is treated as a real resource cost of regulation — delay, discretionary review, litigation risk, compliance — rather than public revenue: it is not rebated to residents and does not enter any budget constraint, consistent with how Glaeser and Gyourko

(2002) measure the zoning tax empirically. This keeps household optimization and every downstream object ($\tilde{V}_{ij}, \Phi_i, N_i, q_i^R$) exactly as derived below, unaffected by the fee except through (4).

The municipality sets $\{t_i^F\}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government acts as a representative of its residents, who face a tradeoff between lower housing costs — which favor a higher fee, pushing land toward residential use — and lower congestion and fewer workers near their homes — which favor a lower fee, leaving land available for commercial use. Formally, municipality $g$ solves

$$\max_{\{t_i^F\}_{i\in S^g}} \; W^g = \sum_{i\in S^g} N_i\,\bar{U}_i \quad \text{s.t.} \quad (4), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $N_i$ is the equilibrium population at location $i$ and $\bar{U}_i = \Gamma\big(1-\frac{1}{\epsilon}\big)\,\Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location $i$, taking other municipalities' fees $\{t_{i'}^F\}_{i'\notin S^g}$ and transportation investment $\{I_{jk}\}$ as given. $\phi(N_i, L_i)$ is a disutility from residing at $i$, capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial\phi}{\partial N_i} > 0, \qquad \frac{\partial\phi}{\partial L_i} > 0,$$

where $N_i$ denotes residential population and $L_i$ denotes employment at location $i$ (defined above). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta\, N_i^a\, L_i^b, \qquad \eta > 0,\; a, b > 0, \tag{6}$$

where $\eta$ is an overall congestion scale and $a, b$ govern the relative importance of residential crowding versus worker influx. The marginal congestion costs are

$$\frac{\partial\phi}{\partial N_i} = \eta a\, N_i^{a-1} L_i^b, \qquad \frac{\partial\phi}{\partial L_i} = \eta b\, N_i^a L_i^{b-1}, \tag{7}$$

both positive. Unlike an additively separable disutility, the two congestion sources are here mutual complements: $\partial^2\phi/\partial N_i \partial L_i = \eta ab\, N_i^{a-1} L_i^{b-1} > 0$, so an additional resident is more burdensome the more workers already commute into $i$, and vice versa. Own-argument convexity, $\partial^2\phi/\partial N_i^2 = \eta a(a-1) N_i^{a-2} L_i^b > 0$, requires $a > 1$, ensuring the first-order condition below has an interior solution.

**Characterizing the optimum.** Since $\bar{H}_i^R = \bar{H}_i - \bar{H}_i^F(t_i^F)$ is a strictly increasing, invertible function of $t_i^F$, it is equivalent — and algebraically simpler — to characterize the fee choice by differentiating the objective with respect to $\bar{H}_i^R$ directly, then translating back to the fee at the end. Substituting $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ and differentiating $\sum_{i\in S^g} N_i \bar{U}_i$ with respect to $\bar{H}_i^R$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (in-movers)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial q_i^R}\frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (rent benefit)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial \phi}\frac{\partial \phi}{\partial N_i}\frac{\partial N_i}{\partial \bar{H}_i^R}}_{\leq 0 \text{ (congestion cost)}} + N_i \underbrace{\left(-\frac{\partial \bar{U}_i}{\partial \phi}\frac{\partial \phi}{\partial L_i}\frac{\partial L_i}{\partial \bar{H}_i^F}\right)}_{\geq 0 \text{ (less worker influx)}} = 0. \tag{8}$$

The first term is positive: a higher fee $t_i^F$ pushes land toward residential use, and the resulting looser residential zoning lowers rents, attracting additional residents from the rest of the metro area; each in-mover contributes $\bar{U}_i$ to the municipality's objective. The second term is positive: lower rents raise effective income for all $N_i$ existing residents. The third term is negative: in-movers raise $N_i$, worsening congestion for all current residents through $\phi$. The fourth term is positive: shifting land from commercial to residential reduces employment $L_i$ and the associated worker congestion. The municipality sets the fee where the benefit of attracting residents and lowering rents is offset by the congestion they impose.

Substituting the Cobb-Douglas marginal congestion costs (7) into the congestion-cost and worker-influx terms makes the interaction between the two channels explicit:

$$\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta a\, N_i^{a-1} L_i^b \frac{\partial N_i}{\partial \bar{H}_i^R} - N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta b\, N_i^a L_i^{b-1} \frac{\partial L_i}{\partial \bar{H}_i^F} = 0. \qquad (9)$$

Because $\partial \bar{U}_i / \partial \phi < 0$ (higher disutility lowers $\tilde{V}_{ij}$, hence $\Phi_i$ and $\bar{U}_i$, for every $j$), the third term is a product of a negative and two non-negative factors and so is $\leq 0$, and the fourth term (after the leading minus sign) is $\geq 0$, matching the signs already established above. The Cobb-Douglas cross-term makes clear that the congestion cost of loosening residential zoning at $i$ is amplified when $i$ already hosts substantial employment $L_i$ ($N_i^{a-1} L_i^b$ rising in $L_i$ for $b > 0$), and symmetrically that the worker-influx benefit of shifting land toward residential use is larger where residential population $N_i$ is already high ($N_i^a L_i^{b-1}$ rising in $N_i$) — a complementarity the additively separable form could not generate.

**Translating back to the fee.** By the chain rule and (5),

$$\frac{dW^g}{dt_i^F} = \frac{dW^g}{d\bar{H}_i^R} \cdot \frac{d\bar{H}_i^R}{dt_i^F}, \qquad \frac{d\bar{H}_i^R}{dt_i^F} = -\frac{1}{g'(\bar{H}_i^R)} \neq 0, \qquad (10)$$

so $dW^g/dt_i^F = 0$ holds if and only if $dW^g/d\bar{H}_i^R = 0$, i.e., if and only if (8) (equivalently (9)) holds at the residential floor space implied by the arbitrage condition (4). Choosing the fee $t_i^F$ therefore characterizes exactly the same interior solution as the underlying quantity choice; it is simply monotonically rescaled by $-1/g'(\bar{H}_i^R) > 0$, which does not change where the derivative vanishes.

$\bar{H}_i^F$ itself can equivalently be read as a *quota*: a cap on how much land may be used commercially. The fee and quota formulations describe the same one-dimensional municipal choice at $i$; the model could equally be stated with a residential-side fee $t_i^R \equiv q_i^R - q_i^F$ (so that $q_i^F = q_i^R - t_i^R$), which is strictly decreasing (rather than increasing) in $\bar{H}_i^R$ but satisfies the same equivalence by the symmetric argument. I take $t_i^F$ — a fee on commercial use — as the primary instrument going forward, since it maps most directly onto observed commercial linkage-fee policy and, together with Glaeser and Gyourko (2002)'s price/construction-cost wedge, onto how zoning restrictiveness is measured empirically.

## 1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{jk}\}_{j,k \in S}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ as given:

$$\max_{\{I_{jk}\}} N \cdot \Gamma\big(1 - \tfrac{1}{\epsilon}\big)\, \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \qquad I_{jk} \geq 0,$$

where $\delta_{jk}$ is the per-unit cost of investment on link $(j,k)$ and $K$ is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. The interior first-order condition for link $(j,k)$ is

$$\frac{N}{\epsilon}\,\Gamma\big(1-\tfrac{1}{\epsilon}\big)\,\Phi^{1/\epsilon-1}\frac{\partial\Phi}{\partial I_{jk}} = \mu\,\delta_{jk}. \tag{11}$$

The left-hand side is the marginal social benefit of investment on link $(j,k)$: lower commuting costs raise $\tilde{V}_{ij}$ for all commuters on that link, increasing $\Phi$ and hence aggregate welfare. The authority equates the marginal benefit per dollar of investment across all active links.

## 1.6 Equilibrium

**Definition 1** (Decentralized Equilibrium)**.** A decentralized equilibrium is a profile

$$\Big(\{\bar{H}_i^{R*},\,\bar{H}_i^{F*}\}_{i\in S},\ \{I_{jk}^*\},\ \{q_i^{R*},\,q_i^{F*},\,w_j^*\}_{i,j\in S}\Big)$$

such that:

1. **Household optimality.** Commuting flows $\{N\pi_{ij}^*\}$ are given by the Fréchet probability formula at prices $(q_i^{R*}, w_j^*)$ and investment $I^*$.
2. **Firm optimality.** Wages $w_j^*$ and commercial rents $q_j^{F*}$ satisfy the firm first-order conditions at $H_j^F = \bar{H}_j^{F*}$ and $L_j^* = N\sum_i \pi_{ij}^*$.
3. **Market clearing.** Residential floor space clears at $q_i^{R*}$ given by (3), commercial floor space is fixed at $H_j^F = \bar{H}_j^{F*}$, and the labor market clears at $L_j^* = N\sum_i \pi_{ij}^*$.
4. **Municipal Nash equilibrium.** Each municipality $g$ sets the fee $\{t_i^{F*}\}_{i\in S^g}$ — equivalently, induces $\{\bar{H}_i^{R*},\bar{H}_i^{F*}\}_{i\in S^g}$ via (10) — to satisfy the first-order condition (8), given $\{\bar{H}_{i'}^{R*},\bar{H}_{i'}^{F*}\}_{i'\notin S^g}$ and $I^*$; no municipality has a profitable deviation.
5. **Transportation authority optimality.** $\{I_{jk}^*\}$ satisfies the first-order condition (11), maximizing aggregate welfare given $\{\bar{H}_i^{R*},\bar{H}_i^{F*}\}$ subject to the budget constraint.

Conditions (iv) and (v) jointly constitute the simultaneous Nash equilibrium between municipalities and the transportation authority: each set of agents takes the other's choices as given and best-responds. The equilibrium fee $t_i^{F*} \equiv q_i^{F*} - q_i^{R*}$ is computed from the equilibrium quantities in this profile, not solved for as a separate object.

## 1.7 Quantification

The household block delivers a log-linear estimating system once the congestion term is handled: writing $w_j - \phi_i = w_j(1-\phi_i/w_j)$ and taking a first-order log approximation, $\ln(w_j-\phi_i) \approx \ln w_j - \phi_i/w_j$. Substituting into $\ln\pi_{ij} = \ln T_i + \ln E_j + \epsilon\ln\tilde{V}_{ij} - \ln\Phi$ with $\tau_{ij} = \delta_{ij}^{\tau} n_{ij}^{\rho}/I_{ij}^{\gamma}$ gives a structural-gravity regression on observed commuting flows $N\pi_{ij}$:

$$\begin{aligned}\ln\pi_{ij} \approx\ & \underbrace{\big[\ln T_i + \epsilon\ln B_i - \epsilon(1-\alpha)\ln q_i^R\big]}_{\text{origin FE}} + \underbrace{\big[\ln E_j + \epsilon\ln w_j\big]}_{\text{destination FE}} - \epsilon\frac{\phi_i}{w_j}\\ & - \epsilon\ln\delta_{ij}^{\tau} - \epsilon\rho\ln n_{ij} + \epsilon\gamma\ln I_{ij} - \ln\Phi, \end{aligned} \tag{12}$$

in the spirit of Ahlfeldt et al. (2015). With origin and destination fixed effects, the coefficients on $\ln n_{ij}$, $\ln I_{ij}$, and $\phi_i/w_j$ identify $-\epsilon\rho$, $\epsilon\gamma$, and $-\epsilon$ respectively. Given $\epsilon$, the origin fixed effect nets out $\ln T_i + \epsilon \ln B_i$ as a residual (observed $q_i^R, \alpha$), the standard model-inversion step.

## 1.8 Data and Calibration

Following Fajgelbaum and Schaal (2020)'s treatment of network and cost fundamentals, parameters split into three groups: estimated from the model's own equations (§Quantification), set from external estimates, and location/link fundamentals recovered as residuals or calibrated directly to observed infrastructure costs.

**Data.** Commuting flows $N\pi_{ij}$ and employment $L_i$: LODES/LEHD origin-destination data. Wages $w_j$ and population $N_i$: ACS. Residential and commercial rents $q_i^R, q_i^F$: county appraisal district records or commercial listing data (CoStar), jointly identifying $t_i^F$ via (4). Land area $\bar{H}_i$: parcel/GIS data. Road and transit network geometry and travel times, for $\delta_{ij}^{\tau}$: GIS road network and transit schedules. Historical investment $I_{jk}$ and per-unit construction costs $\delta_{jk}$: NTD and state DOT capital programs. Total budget $K$: observed metro-area transportation capital spending over the relevant period.

**Estimated from model equations.** $\epsilon, \rho$, and the commuting-cost investment elasticity in (12); $(\eta, a, b)$ from the congestion regression; $t_i^F$ is not estimated at all, but read directly off rent data via (4).

# References


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